

Portfolio Website — Comprehensive Technical Audit

Audit Date: July 4, 2026
Auditor: Senior Full-Stack Engineer & Solutions Architect
Repository: e:\WORKS\portfolio

1. Tech Stack & Dependency Audit

Core Stack Identified

Technology	Version	Latest (Jul 2026)	Status
Next.js	^16.1.1	~16.x	✅ Reasonably current
React	19.2.0	19.x	✅ Current
TypeScript	^5	5.x	✅ Current
Tailwind CSS	^4	4.x	✅ Current (v4 architecture)
Node.js	Not pinned	22 LTS / 24	⚠️ No .nvmrc / engines field

Dependency-by-Dependency Analysis

Production Dependencies

Package	Version	Verdict
@tailwindcss/typography	^0.5.19	⚠️ Version mismatch — This is the v3-era plugin. Tailwind v4 uses @plugin directive (which you are using in CSS), but the npm package version should be aligned with the Tailwind v4 plugin system.
@types/nodemailer	^7.0.4	⚠️ Misplaced — Type packages belong in devDependencies, not dependencies.
clsx	^2.1.1	✅ Fine, but unused — no imports found anywhere in the codebase.
framer-motion	^12.23.25	✅ Current
gray-matter	^4.0.3	✅ Stable, no major changes
lucide-react	^0.555.0	✅ Current
next	^16.1.1	✅ Current
next-mdx-remote	^5.0.0	✅ Current
next-themes	^0.4.6	✅ Current
nodemailer	^7.0.12	✅ Current
react / react-dom	19.2.0	✅ Current, but pinned (no ^) — intentional or oversight?
tailwind-merge	^3.4.0	⚠️ Unused — no imports of tailwind-merge found anywhere in the codebase.

Dev Dependencies

Package	Version	Verdict
@tailwindcss/postcss	^4	✅ Correct for TW v4
@types/node	^20	⚠️ Should be ^22 to match Node 22 LTS
@types/react / @types/react-dom	^19	✅ Matches React 19

Package	Version	Verdict
eslint	^9	✅ Current
eslint-config-next	^16.1.1	✅ Matches Next.js version
tailwindcss	^4	✅ Current
typescript	^5	✅ Current

Key Dependency Issues

 Warning

Unused packages detected: `clsx` and `tailwind-merge` are installed but never imported. These add ~15KB to `node_modules` and represent dead weight.

 Important

Missing critical tooling: No Prettier, no Husky/lint-staged, no test framework (Jest/Vitest/Playwright), no `@next/bundle-analyzer` . For a 2026 portfolio, these are expected standards.

2. Architecture & Directory Structure

High-Level Structure Map

```
portfolio/
├── .env                                # ⚠️ SECURITY CONCERN (see §5)
├── .gitignore
├── docs/
│   └── Portfolio SRS.pdf              # Spec doc (not referenced by app)
├── eslint.config.mjs
├── next.config.ts
├── package.json
├── postcss.config.mjs
├── public/
│   ├── file.svg                      # 🔴 Default Next.js scaffolding (unused)
│   ├── globe.svg                     # 🔴 Default Next.js scaffolding (unused)
│   ├── next.svg                      # 🔴 Default Next.js scaffolding (unused)
│   ├── vercel.svg                    # 🔴 Default Next.js scaffolding (unused)
│   └── window.svg                    # 🔴 Default Next.js scaffolding (unused)
├── src/
│   ├── app/
│   │   ├── globals.css
│   │   ├── layout.tsx                # Root layout
│   │   ├── page.tsx                 # Home page
│   │   ├── about/page.tsx
│   │   ├── blog/
│   │   │   ├── page.tsx
│   │   │   └── [slug]/page.tsx
│   │   ├── contact/
│   │   │   ├── page.tsx
│   │   │   └── actions.ts            # Server Action (email)
│   │   └── projects/
│   │       ├── page.tsx
│   │       └── [id]/page.tsx
│   ├── components/                   # 🔴 Flat – no sub-grouping
│   │   ├── CallToAction.tsx
│   │   ├── ContactForm.tsx
│   │   ├── Footer.tsx
│   │   ├── Hero.tsx
│   │   ├── Navbar.tsx
│   │   ├── ProjectCard.tsx
│   │   ├── ProjectDetails.tsx
│   │   ├── TechStack.tsx
│   │   ├── ThemeToggle.tsx
│   │   └── theme-provider.tsx
│   ├── content/
│   │   └── blog/
│   │       └── why-your-ecommerce-crashes.mdx
│   └── lib/
│       ├── blog.ts                   # Blog file reader (fs-based)
│       └── data.ts                   # 🔴 MEGA FILE – 220 lines of hardcoded data
└── tsconfig.json
```

Architectural Assessment

Aspect	Finding	Severity
Pattern	Next.js App Router (correct)	✓
Component organization	Flat /components directory — no grouping by feature or domain (layout, ui, sections)	⚠️ Medium

Aspect	Finding	Severity
Data layer	All site data hardcoded in a single [data.ts](file:///e:/WORKS/portfolio/src/lib/data.ts) file (220 lines). No CMS, no API, no JSON files, no separation by entity.	🔴 High
Type safety	any type used for experience/education mapping in [about/page.tsx](file:///e:/WORKS/portfolio/src/app/about/page.tsx#L27) (lines 27, 53). No exported Project type despite complex project objects.	⚠️ Medium
Naming inconsistency	theme-provider.tsx uses kebab-case while all other components use PascalCase.	⚠️ Low
Build artifacts committed	.next/ and out/ directories exist locally (gitignored, but cluttering workspace). tsconfig.tsbuildinfo present.	⚠️ Low

⚠️ Caution

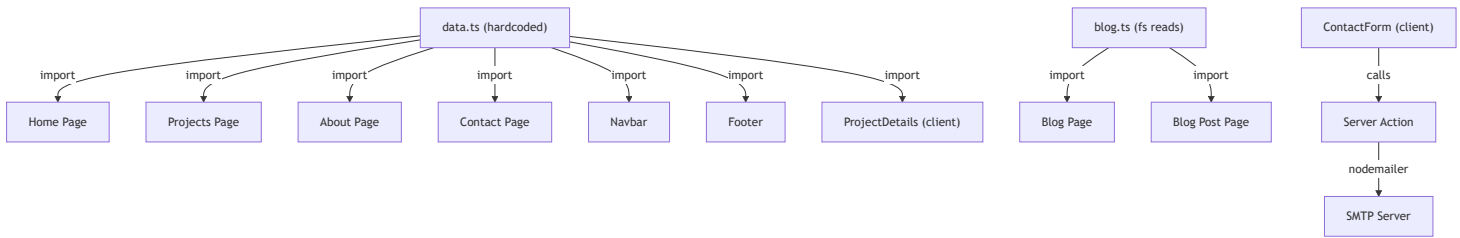
Anti-pattern: God Data File — [data.ts](file:///e:/WORKS/portfolio/src/lib/data.ts) contains `SITE_CONFIG`, `HERO_CONTENT`, `ABOUT_CONTENT`, `TECH_STACK`, and `PROJECTS` (9 project objects with full details). This is the single biggest maintainability bottleneck. Any content update requires editing source code and redeploying.

3. Routing & Data Flow

Route Map

Route	File	Type	Data Source
/	[page.tsx](file:///e:/WORKS/portfolio/src/app/page.tsx)	Server Component (SSG)	PROJECTS from data.ts
/about	[about/page.tsx](file:///e:/WORKS/portfolio/src/app/about/page.tsx)	Server Component (SSG)	ABOUT_CONTENT from data.ts
/projects	[projects/page.tsx](file:///e:/WORKS/portfolio/src/app/projects/page.tsx)	Server Component (SSG)	PROJECTS from data.ts
/projects/[id]	[projects/[id]/page.tsx](file:///e:/WORKS/portfolio/src/app/projects/%5Bid%5D/page.tsx)	Server Component (SSG)	PROJECTS from data.ts → ProjectDetails client component
/blog	[blog/page.tsx](file:///e:/WORKS/portfolio/src/app/blog/page.tsx)	Server Component (SSG)	getSortedPostsData() from blog.ts (fs reads)
/blog/[slug]	[blog/[slug]/page.tsx](file:///e:/WORKS/portfolio/src/app/blog/%5Bslug%5D/page.tsx)	Server Component (SSG)	getPostData() from blog.ts + MDXRemote
/contact	[contact/page.tsx](file:///e:/WORKS/portfolio/src/app/contact/page.tsx)	Server Component + Client Form	Server Action <code>sendEmail()</code>

Data Flow Analysis



Issues Identified

- No state management needed** — The app is essentially static. No Redux, no Context (except `ThemeProvider`), no API calls. This is architecturally sound for a portfolio but limits future extensibility.
- "use client" overuse** — 8 of 10 components are client components. `[Hero.tsx]`(`file:///e:/WORKS/portfolio/src/components/Hero.tsx`), `[TechStack.tsx]`(`file:///e:/WORKS/portfolio/src/components/TechStack.tsx`), and `[ProjectCard.tsx]`(`file:///e:/WORKS/portfolio/src/components/ProjectCard.tsx`) are client components *solely* for Framer Motion animations. These could use the `motion` component from `framer-motion/m` (server-compatible) or be refactored to minimize the client boundary.
- ProjectDetails calls notFound() in a client component** — `[ProjectDetails.tsx L17]`(`file:///e:/WORKS/portfolio/src/components/ProjectDetails.tsx#L17`): `notFound()` from `next/navigation` is called inside a client component. While this works, it's an anti-pattern. The 404 logic should live in the server component (`[projects/[id]/page.tsx]`(`file:///e:/WORKS/portfolio/src/app/projects/%5Bid%5D/page.tsx`)) before the data is passed down.
- No loading/error states** — Zero `loading.tsx`, `error.tsx`, or `not-found.tsx` files anywhere. Users will see raw errors on failure.
- Blog system is fragile** — `[blog.ts]`(`file:///e:/WORKS/portfolio/src/lib/blog.ts`) uses synchronous `fs.readFileSync` and `fs.readdirSync`. Only 1 blog post exists. The system works but has no pagination, search, or category filtering.
- No generateMetadata on project detail pages** — The `/projects/[id]` route has no dynamic metadata generation, hurting SEO.

4. UI/UX & Styling Analysis

Styling Solution

- Tailwind CSS v4** with `@tailwindcss/postcss` plugin
- Custom CSS variables** in `[globals.css]`(`file:///e:/WORKS/portfolio/src/app/globals.css`) using a `shadcn/ui`-inspired HSL token system
- Dark mode** via `next-themes` with class strategy
- Fonts:** `Inter (sans)` + `JetBrains Mono (mono)` via `next/font`

Design System Assessment

Aspect	Rating	Notes
Color palette	⚠️ Flat	Pure black/white theme (<code>0 0% 0% / 0 0% 100%</code>). Zero brand color, zero accent color, zero personality. Every <code>--primary</code> is literally black or white.
Typography	✅ Good	Inter + JetBrains Mono is a strong pairing. Font optimization via <code>next/font</code> .
Spacing	⚠️ Inconsistent	Mix of container <code>mx-auto px-4</code> and <code>max-w-7xl mx-auto px-4</code> . The layout already has <code>max-w-7xl mx-auto px-4</code> in the root layout, but pages add it again → double-constrained width .
Animations	⚠️ Basic	Only Framer Motion fade-in (<code>opacity: 0→1</code> , <code>y: 20→0</code>). Same animation on every component. No scroll-triggered reveals, no hover micro-interactions beyond color changes.
Responsive	✅ Adequate	Mobile hamburger menu, responsive grids. No tablet-specific breakpoints.
Accessibility	⚠️ Partial	<code>aria-label</code> on some buttons, <code>sr-only</code> on theme toggle. Missing: skip-to-content link, focus-visible styles, ARIA landmarks, color contrast issues with muted-foreground on dark mode.

Component Reusability Issues

1. **No shared Button component** — Button styles are duplicated inline across [Hero.tsx](file:///e:/WORKS/portfolio/src/components/Hero.tsx#L28), [ContactForm.tsx](file:///e:/WORKS/portfolio/src/components/ContactForm.tsx#L104-L120), [ProjectDetails.tsx](file:///e:/WORKS/portfolio/src/components/ProjectDetails.tsx#L65-L84). The class string `inline-flex items-center justify-center px-6 py-3 rounded-md bg-primary text-primary-foreground font-medium hover:bg-primary/90 transition-color` is repeated **3+ times**.
2. **No shared Input component** — Form input styles are duplicated across the name, email, and message fields in [ContactForm.tsx](file:///e:/WORKS/portfolio/src/components/ContactForm.tsx#L54-L89).
3. **Tag/Badge rendering duplicated** — Tag pill styles appear in [ProjectCard.tsx](file:///e:/WORKS/portfolio/src/components/ProjectCard.tsx#L50-L57), [ProjectDetails.tsx](file:///e:/WORKS/portfolio/src/components/ProjectDetails.tsx#L49-L56), [blog/page.tsx](file:///e:/WORKS/portfolio/src/app/blog/page.tsx#L30-L34), and [blog/[slug]/page.tsx](file:///e:/WORKS/portfolio/src/app/blog/%5Bslug%5D/page.tsx#L49-L53).
4. **Section header pattern duplicated** — The `h1 + subtitle paragraph` pattern is copy-pasted across About, Blog, Projects, and Contact pages.
5. **Hardcoded strings in components** — "Start a Project", "View Portfolio", "Need a high-performance solution?", "Message sent successfully! (Demo mode)" are all hardcoded in JSX instead of being in a constants file or i18n system.
6. **No image assets** — The entire portfolio has **zero project screenshots, zero profile photo, zero visual assets**. Public folder contains only default Next.js SVGs. For a portfolio site this is a critical gap.

Custom CSS Utilities

Two custom classes defined but **barely used**:

- `.font-mono-accent` — **never used** in any component
- `.glass-panel` — **never used** in any component

5. Technical Debt & Security

● CRITICAL: Security Vulnerabilities

⚠ Caution

Exposed SMTP Credentials in `.env` file

The `[.env](file:///e:/WORKS/portfolio/.env)` file contains **plaintext email credentials**:

```
EMAIL_PASSWORD = 'oekm opvi tuns eoxg'
SMTP_PASS=egvzyngmx1oksliv
```

While `.env` is gitignored and NOT tracked in git (confirmed), the file contains:

- **Two separate sets of email credentials** (lines 1-2 vs lines 5-8) — suggesting abandoned credentials that were never cleaned up
- `EMAIL_USERNAME / EMAIL_PASSWORD` on lines 1-2 are **completely unused** by any code — they're dead credentials sitting on disk
- App passwords from Google accounts are exposed in plaintext

⚠ Warning

XSS Vulnerability in Contact Server Action

In `[actions.ts L51-L59](file:///e:/WORKS/portfolio/src/app/contact/actions.ts#L51-L59)`, user-submitted `name`, `email`, and `message` values are interpolated directly into an HTML template string **without sanitization**:

```
html: `<p><strong>Name:</strong> ${name}</p>`
```

An attacker could submit `<script>alert('xss')</script>` as a name. While this primarily affects the email client rendering, it's still unsanitized user input in HTML context.

 Warning

No rate limiting on contact form — The `sendEmail` server action has zero rate limiting. A bot could spam the endpoint, exhausting SMTP quotas and potentially getting the email account flagged.

 Warning

No input validation — `[actions.ts](file:///e:/WORKS/portfolio/src/app/contact/actions.ts#L13-L15)`: Email format is not validated server-side. Only checks for empty strings with `if (!name || !email || !message)`. No length limits, no email regex, no content sanitization.

 Technical Debt Items

Item	Location	Severity
Dead CSS classes	<code>.font-mono-accent</code> , <code>.glass-panel</code> in <code>[globals.css](file:///e:/WORKS/portfolio/src/app/globals.css#L96-L103)</code>	Low
Unused dependencies	<code>clsx</code> , <code>tailwind-merge</code> in <code>[package.json](file:///e:/WORKS/portfolio/package.json#L14-L24)</code>	Low
Dead env variables	<code>EMAIL_USERNAME</code> , <code>EMAIL_PASSWORD</code> in <code>[.env](file:///e:/WORKS/portfolio/.env#L1-L2)</code>	Medium
any type usage	<code>exp: any</code> and <code>edu: any</code> in <code>[about/page.tsx](file:///e:/WORKS/portfolio/src/app/about/page.tsx#L27-L53)</code>	Medium
Unsafe type assertion	<code>as typeof project & {...}</code> in <code>[ProjectDetails.tsx L24-29](file:///e:/WORKS/portfolio/src/components/ProjectDetails.tsx#L24-L29)</code>	Medium
initialState declared but unused	<code>[ContactForm.tsx L9-12](file:///e:/WORKS/portfolio/src/components/ContactForm.tsx#L9-L12)</code>	Low
Cpu icon imported but unused	<code>[data.ts L1](file:///e:/WORKS/portfolio/src/lib/data.ts#L1)</code> — <code>Cpu</code> is imported from <code>lucide-react</code> but never used	Low
console.error in production	<code>[actions.ts L25](file:///e:/WORKS/portfolio/src/app/contact/actions.ts#L25)</code> and <code>[L65](file:///e:/WORKS/portfolio/src/app/contact/actions.ts#L65)</code>	Low
Placeholder URLs everywhere	All <code>codeUrl</code> values in <code>[data.ts](file:///e:/WORKS/portfolio/src/lib/data.ts)</code> point to <code>https://github.com/sahadcse</code> with <code>// Placeholder</code> comments, and social links are marked <code>// Placeholder</code>	Medium
Default README	[README.md](file:///e:/WORKS/portfolio/README.md) is the unmodified <code>create-next-app</code> boilerplate	Low
Default public assets	<code>file.svg</code> , <code>globe.svg</code> , <code>next.svg</code> , <code>vercel.svg</code> , <code>window.svg</code> are the default Next.js scaffold files, never referenced	Low
Commented-out config	<code>// output: "export"</code> in <code>[next.config.ts](file:///e:/WORKS/portfolio/next.config.ts#L4)</code>	Low
Images unoptimized	<code>images: { unoptimized: true }</code> in <code>[next.config.ts](file:///e:/WORKS/portfolio/next.config.ts#L5-L7)</code> disables Next.js Image Optimization entirely	Medium
Inconsistent import paths	<code>[layout.tsx L5](file:///e:/WORKS/portfolio/src/app/layout.tsx#L5)</code> : Footer is imported as <code>"../components/Footer"</code> (relative) while all other imports use <code>"@/components/..."</code> (alias)	Low
No rel="noopener" on some external links	<code>[ProjectCard.tsx](file:///e:/WORKS/portfolio/src/components/ProjectCard.tsx#L34-L41)</code> and <code>[ProjectDetails.tsx](file:///e:/WORKS/portfolio/src/components/ProjectDetails.tsx#L65-L84)</code> external links lack <code>rel="noopener noreferrer"</code>	Low
Double max-width constraint	Root layout has <code>max-w-7x1 mx-auto px-4</code> , and pages add <code>container mx-auto px-4</code> or <code>max-w-7x1 mx-auto px-4</code> again inside	Low
Blog date format	Blog dates display as raw ISO strings (e.g., "2024-11-28") instead of human-readable format	Low

Missing Production Essentials

- ❌ No robots.txt
- ❌ No sitemap.xml (or sitemap.ts generation)
- ❌ No Open Graph / Twitter Card metadata
- ❌ No manifest.json / PWA support
- ❌ No structured data (JSON-LD) for SEO
- ❌ No analytics integration
- ❌ No error monitoring (Sentry, etc.)
- ❌ No Content Security Policy headers
- ❌ No X-Frame-Options , X-Content-Type-Options security headers

6. Migration & Upgrade Suggestions

✅ KEEP (Working Well)

- ✅ **Next.js 16 + App Router** — Modern, correct architecture
- ✅ **React 19** — Current
- ✅ **TypeScript** — Correct choice
- ✅ **Tailwind CSS v4** — Current, with PostCSS plugin
- ✅ **next-themes dark mode** — Clean implementation
- ✅ **Font strategy** — Inter + JetBrains Mono via next/font is solid
- ✅ **HSL CSS variable token system** — Good foundation for theming
- ✅ **Server Actions for contact form** — Correct Next.js pattern
- ✅ **MDX blog system** — Good foundation, needs expansion
- ✅ **generateStaticParams** for static generation — Correct

🔧 REFACTOR (Needs Improvement)

- ☐ **Extract a proper type system** — Create src/types/ with Project , Experience , Education , BlogPost , SiteConfig interfaces. Eliminate all any usage.
- ☐ **Split data.ts into separate files** — src/data/projects.ts , src/data/experience.ts , src/data/site-config.ts , or better yet, move to MDX/JSON content files.
- ☐ **Create shared UI components** — Button , Input , Badge/Tag , SectionHeader , Container components to eliminate duplication.
- ☐ **Organize components by domain** — src/components/layout/ , src/components/ui/ , src/components/sections/ .
- ☐ **Add loading/error/not-found UI** — loading.tsx , error.tsx , not-found.tsx at minimum at the root layout level.
- ☐ **Move notFound() logic out of client component** — Handle in [projects/[id]/page.tsx](file:///e:/WORKS/portfolio/src/app/projects/%5Bid%5D/page.tsx) server component.
- ☐ **Add generateMetadata to project detail pages** — For dynamic SEO.
- ☐ **Sanitize contact form HTML** — Use a template library or escape user input.
- ☐ **Add rate limiting** — Use server-side rate limiting on the contact form action.
- ☐ **Fix double max-width constraint** — Choose one approach: either the layout constrains width OR each page does, not both.
- ☐ **Reduce client component boundary** — Wrap only the animated portions in "use client" , or use Framer Motion's server-compatible APIs.

● REPLACE (Needs Complete Overhaul)

- ☐ **Design system / Color palette** — Replace the monochrome black/white palette with a modern, branded color scheme with actual accent colors, gradients, and visual personality.
- ☐ **Visual assets** — Add project screenshots, a profile photo, custom icons. A portfolio with zero imagery is fundamentally incomplete.
- ☐ **README.md** — Replace the default create-next-app README with actual project documentation.
- ☐ **Public folder** — Delete all default Next.js SVGs. Add proper favicon set (.ico , apple-touch-icon.png , etc.), OG images, and any needed static assets.
- ☐ **Contact form security** — Add proper validation (Zod), rate limiting, CSRF protection, and HTML sanitization.
- ☐ **.env cleanup** — Remove dead EMAIL_USERNAME / EMAIL_PASSWORD variables. Rotate the SMTP credentials since they've been on disk in plaintext.

NEW

ADD (Missing for 2026 Standards)

- ☐ **SEO:** `sitemap.ts` , `robots.ts` , Open Graph images, JSON-LD structured data
- ☐ **Security headers:** via `next.config.ts` → `headers()` function or middleware
- ☐ **Analytics:** Vercel Analytics, Google Analytics, or Plausible
- ☐ **Error monitoring:** Sentry or similar
- ☐ **Testing:** At minimum Vitest for unit tests, Playwright for E2E
- ☐ **CI/CD:** GitHub Actions for lint, type-check, build, test on PRs
- ☐ **Code quality:** Prettier config, Husky + lint-staged for pre-commit hooks
- ☐ **Performance:** Enable Next.js Image Optimization (remove `unoptimized: true`), add `@next/bundle-analyzer`
- ☐ **Resume/CV download** — Standard portfolio feature, completely absent
- ☐ **Animations upgrade** — Scroll-triggered reveals, page transitions, micro-interactions, cursor effects for a premium feel
- ☐ **Blog enhancements** — More than 1 post, reading time calculation, table of contents, related posts, social sharing

Summary Scorecard

Category	Score	Verdict
Tech Stack Freshness	8/10	Stack is modern, minor dependency hygiene issues
Architecture	5/10	App Router is correct, but data layer is a monolith
Routing & Data Flow	6/10	Routes are clean, but client boundary is too wide
UI/UX & Design	3/10	Functional but visually flat, no brand identity, no images
Code Quality	5/10	Works but has dead code, <code>any</code> types, and no tests
Security	3/10	Exposed credentials, no validation, no rate limiting, no security headers
SEO & Performance	4/10	Basic metadata exists, but missing sitemap, OG tags, image optimization
Overall	4.9/10	Solid foundation buried under neglect — needs a focused rewrite, not a patch job



Tip

Recommended approach: This codebase has a solid architectural skeleton (Next.js App Router, TypeScript, Tailwind v4). Rather than scrapping everything, I recommend a **systematic rewrite in place** — keeping the routing structure and build tooling while overhauling the design system, component library, data layer, and security posture. The highest-ROI work is: (1) new visual design with real imagery, (2) proper component library, (3) security hardening, (4) SEO completion.